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The Editor-in-Chief
IEEE Transactions on Medical Imaging

Dear Editor-in-Chief,

We are pleased to submit our manuscript, “FreeReg: Free-Coordinate Registration of Arbitrary 3D Models to Arbitrary Image Streams for Augmented and Mixed Reality,” for consideration as a Regular Paper in *IEEE Transactions on Medical Imaging*.

Augmented-reality guidance in interventional medicine hinges on a registration problem: aligning a preoperative 3D model with the live endoscopic view. Deployed systems either depend on external tracking hardware (electromagnetic navigation) or assume calibrated, rigid, per-frame conditions that fail under uncalibrated optics, tissue deformation, and temporal jitter. We formulate this as free-coordinate estimation of the projection mapping among model, camera, and pixel frames, in which intrinsics, depth, rigid pose, and deformation are each optional and degradable. FreeReg combines a multi-source correspondence engine with contour-gated arbitration, an observability-guided non-rigid estimator confined to the spectrally observable subspace, a manifold $SE(3)$ Kalman filter with innovation gating, and a co-anchored registration-rendering field whose photometric residual serves as an online, label-free validation signal.

One and the same engine is validated across rigid CAD (BOP Linemod/T-LESS), deformable tissue (real-colon FEM), and clinical CT-to-bronchoscopy and SCARED endoscopy. It matches or exceeds state of the art where such methods apply (BOP AR 0.848 vs FoundationPose 0.847; T-LESS ADD-S 0.921) and remains correct where they collapse out-of-distribution: on clinical endoscopy, FoundationPose reaches 104 mm and Endo3R 34 mm error, while FreeReg reaches 4.23 mm mean target registration error at 30–80 FPS—within the accuracy range of electromagnetic navigation, without any external tracking hardware. Our observability analysis further quantifies which deformation directions a given sensing condition can and cannot recover, and shows that a second view nearly doubles the recoverable subspace (23% to 41%).

We believe this work is well suited to IEEE TMI. Registration is a recognized bottleneck in image-guided intervention, and FreeReg addresses it with a camera-only, calibration-free approach whose accuracy, robustness, and explicit reliability limits are directly relevant to the clinical adoption of surgical AR—while its free-coordinate formulation and observability theory are of broad methodological interest to the medical-imaging community.

This manuscript is original, has not been published previously, and is not under consideration elsewhere. All authors have read and approved the submission and declare no conflict of interest. The clinical data are de-identified and used under the institutional data-governance process of Henan Provincial People’s Hospital (see Ethics Statement). The full pipeline and evaluation scripts are publicly released at <https://github.com/liranyang123456-commits/FreeReg> for reproducibility.

Thank you for your consideration. We look forward to your response.

Sincerely,

Ranyang Li, Ph.D.
Corresponding author, on behalf of
all authors